

MUSFIQUR RAHMAN

PhD Candidate in Software Engineering | AI/ML Engineer & Researcher | Data & AI Educator

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SUMMARY

PhD candidate with 8+ years of combined research and industry experience spanning AI/ML engineering, applied research, and technical education. Builds and ships production LLM systems with strong engineering discipline (CI/CD, automated testing, reproducible evaluation), and has six years of experience designing and delivering data science and AI curriculum, training 1000+ professionals. Published at ICSE, MSR, EASE, and EMSE with 100+ citations. Open to ML/AI Engineering, Research Engineering, Data Science, and Curriculum/Education roles.

TECHNICAL EXPERTISE

Languages: Python (expert), Java, C/C++, SQL, Bash

ML/AI: LLMs, NLP, Transformers, PyTorch, CatBoost, Scikit-learn, Deep Learning, Time-Series Analysis

LLM Stack: HuggingFace, CodeBERT, Prompt Engineering, RAG, Agentic Workflows, Model Evaluation

Infrastructure: Docker, GCP, AWS, Azure ML, Git, GitHub Actions (CI/CD), PostgreSQL, BigQuery

Instructional: Curriculum design, instructional scaffolding, learner-centric content review, instructor mentorship

PRODUCTION SYSTEMS & KEY PROJECTS

SIEVE — End-to-End LLM Detection System

Sep 2025 – Present

[sieve-tool.streamlit.app](#) | [github.com/mrsumitbd/sieve](#) | [HuggingFace: mrahman2025/sieve-llm-classifier](#)

- Independently designed and shipped an end-to-end ML system: data pipeline, multi-language extraction, transformer-based classification, and live production deployment
- CodeBERT classifier achieving $F1=0.9478$, $AUROC=0.9902$ across 114K samples spanning 9 LLMs; 273-test CI suite at 80% coverage

OpenClassGen & RealClassEval — LLM Evaluation Framework

Apr 2025 – Present

- Built large-scale evaluation framework benchmarking GPT-o4-mini, Claude-4-Sonnet, and Qwen-3-Coder on real-world code generation; curated 324,843-sample dataset released on HuggingFace; published at EASE 2026

LLM Code Detection System

2023 – Present

- Built and evaluated classifier using 44 engineered features; ran systematic ablations and adversarial robustness testing across 4 LLMs and multiple threat conditions

PROFESSIONAL EXPERIENCE

Enterprise Solutions Architect & AI Systems Evaluator

Nov 2025 – May 2026

General Assembly, Remote

- Performed systematic testing and QA of GAIA, GA's AI agent system for automated course content generation; evaluated AI-generated instructional content for pedagogical quality
- Designed AI learning solutions for enterprise clients including Adobe and Diageo; developed L0–L4 content scoping framework

Subject Matter Expert & Lead Educator, Data and AI

Feb 2020 – Nov 2025

General Assembly, Remote

- Designed and delivered curriculum across three data science and AI bootcamp programs (Data Science, Data Analytics, Data Engineering); managed content development lifecycle and deadlines
- Sourced, recruited, and mentored instructors; trained 1000+ professionals across 50+ cohorts spanning Data Science, GenAI Applications, and AI for Healthcare
- Reviewed content from a learner-first perspective, ensuring technical accuracy and pedagogical effectiveness; iterated based on learner feedback

Artificial Intelligence Engineer

Feb 2018 – Feb 2020

Pentavere (Health Tech), Toronto, ON

- Built and deployed production clinical NLP models processing millions of unstructured health records
- Designed end-to-end ML pipelines for automated healthcare data extraction and analysis

Data Scientist (Part-time)

May 2023 – Jan 2024

SGS, Remote

- Performed time-series analysis for financial forecasting (ARIMA, SARIMA, SARIMAX, FB Prophet) on Azure ML platform

RESEARCH EXPERIENCE

PhD Researcher – LLMs for Software Engineering <i>DAS Lab, Concordia University (Advisor: Prof. Emad Shihab), Montréal, QC</i>	<i>May 2022 – Present</i>
• Researching LLM code generation, detection, and evaluation methodology; building and deploying production ML systems as part of research • Mechanistic interpretability: tracing how transformer-based models retrieve and apply information across layers and attention heads	
M.Sc. Researcher – Code Naturalness & Machine Translation <i>CESEL Lab, Concordia University (Advisor: Prof. Peter C. Rigby), Montréal, QC</i>	<i>Sep 2015 – Apr 2018</i>
• Applied NLP to analyze large-scale code corpora; published at ICSE 2019 and MSR 2019	

SELECTED PUBLICATIONS

- **M. Rahman** et al. “OpenClassGen: A Large-Scale Corpus of Real-World Python Classes for LLM Research.” *EASE 2026*. DOI: 10.1145/3816483.3816547.
- **M. Rahman** et al. “Automatic Detection of LLM-Generated Code: Comparative Study Across Function and Class Granularities.” arXiv:2409.01382.
- **M. Rahman** et al. “The Impact of Environment Configurations on the Stability of AI-Enabled Systems.” *EASE 2025*.
- **M. Rahman**, D. Palani, P. C. Rigby. “Natural Software Revisited.” *ICSE 2019*.
100+ citations | ICSE, MSR, EASE, EMSE | [Google Scholar](#)

EDUCATION

PhD in Software Engineering Concordia University, Montréal, QC	<i>May 2022 – Sep 2026 (Expected)</i> GPA: 4.30/4.30
M.Sc. in Computer Science Concordia University, Montréal, QC	<i>Sep 2015 – Apr 2018</i> GPA: 3.90/4.30
B.Sc. in Computer Science and Engineering East West University, Dhaka, Bangladesh	<i>Jan 2011 – Dec 2014</i> GPA: 3.62/4.00

AWARDS & LEADERSHIP

- **Concordia University Doctoral Fellowship** (2022–2026) – Full PhD funding for research excellence
- **Campaign for Concordia Graduate Award** (2022)
- **Academic Service:** PC member/reviewer for ICSME 2026, ICSE-SRC 2026, MSR 2024/2026, TechDebt 2026, CAIN 2026, TOSEM, EMSE